

WHICH LATTICE MINIMISES A STEEP THREE-BODY POWER-LAW ENERGY?

A MAX-MIN TRIANGLE PROBLEM

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ABSTRACT. For a lattice $\Lambda \subset \mathbb{R}^d$ of unit covolume consider the three-body energy $T_\nu(\Lambda) = \sum_{x,y \in \Lambda \setminus \{0\}, x \neq y} (|x||y||x-y|)^{-\nu}$, i.e. the sum over all lattice triangles through the origin of the product of the side lengths to the power $-\nu$. In $d = 2$ this is, up to normalisation, the two-loop modular graph function $C_{s,s,s}(\tau)$ with $s = \nu/2$. We prove that as $\nu \rightarrow \infty$ every family of minimisers of T_ν converges to the rectangular lattice with aspect ratio $\sqrt{(\sqrt{17}-1)/2}$; in particular neither the hexagonal nor the square lattice is optimal for large ν . The proof reduces the limit to a max-min problem for the smallest triangle product of a lattice, which we solve exactly. In $d = 3$ we prove, by a computer-assisted argument (branch and bound over Minkowski-reduced Gram matrices plus an interval-arithmetic local lemma), that the body-centred cubic lattice is the unique maximiser of the smallest triangle product, so that minimisers of T_ν converge to BCC. For finite exponents we prove, again computer-assisted, that $C_{2,2,2}$, $C_{3,3,3}$ and $C_{4,4,4}$ attain their minimum only at $\tau = i$ (square lattice), whereas $C_{1,1,1}$ is minimal at $e^{i\pi/3}$. We complement this with a numerical phase diagram for finite ν (hexagonal $\rightarrow$ square $\rightarrow$ rectangular in $d = 2$, with a certified enclosure of the first transition) and with numerical results on hcp and on mixed pair plus three-body energies.

1. INTRODUCTION

Lattice energy minimisation for *pair* interactions is a classical subject; for completely monotone pair potentials in the plane the hexagonal lattice is optimal [1, 2, 3, 4, 5], and in higher dimensions universal optimality has been proved in dimensions 8 and 24 [6]. Much less is known for genuine many-body energies. Here we study the simplest one, a product-form three-body power law

$$(1) \quad T_\nu(\Lambda) = \sum_{\substack{x,y \in \Lambda \setminus \{0\} \\ x \neq y}} (|x||y||x-y|)^{-\nu},$$

which converges for $\nu > 2d/3$. This sum (under the name three-body zeta function $\zeta_\Lambda^{(3)}(\nu, \nu, \nu)$) appears as the radial part of the Axilrod–Teller–Muto energy at $\nu = 3$ in [7], where it was evaluated for selected lattices and along the Bain path; efficient algorithms are given in [8, 9]. The reduction of a steep-decay limit to a packing-type problem is standard for pair energies [11]; three-point max-min problems have been studied on the sphere [12], and the criticality of symmetric lattices is automatic [13]. We stress that (1) is a *model* energy: physical three-body dispersion carries an angular factor, and a pure three-body energy at fixed density does not describe a material. The motivation here is mathematical, see Section 4: in $d = 2$, T_{2s} is a modular graph function.

What is automatic and what is not. T_ν is invariant under isometries, so on the modular surface the hexagonal point $\rho = e^{i\pi/3}$ and the square point i (elliptic fixed points) are always critical points. The content of the results below is *global* minimality and the location of the transitions.

Main result. Let

$$a_* = \left(\frac{1+\sqrt{17}}{8}\right)^{1/4}, \quad P_* = 2a_*^3 = \sqrt{a_*^2 + a_*^{-2}} \approx 1.4317345, \quad y_\infty = a_*^{-2} = \sqrt{\frac{\sqrt{17}-1}{2}} \approx 1.2496211,$$

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and let $\Lambda_* = a_*\mathbb{Z} \times a_*^{-1}\mathbb{Z}$ (rectangular, aspect ratio y_∞ , unit covolume).

Theorem 1. *Let $d = 2$. For every $\nu > 4/3$ the energy T_ν attains its minimum on the space of unit-covolume lattices. If Λ_ν is any minimiser, then $\Lambda_\nu \rightarrow \Lambda_*$ up to isometries as $\nu \rightarrow \infty$, and $\lim_{\nu \rightarrow \infty} (\min T_\nu)^{1/\nu} = 1/P_*$. Equivalently (Section 4), minimisers τ_s of the modular graph function $C_{s,s,s}$ on the fundamental domain satisfy $\tau_s \rightarrow i y_\infty$ as $s \rightarrow \infty$.*

Theorem 1 follows from an exact solution of the following max-min problem.

Definition 2. For a lattice Λ let $\pi(p, q, r) = |p - q| |q - r| |r - p|$ and $P(\Lambda) = \min\{\pi(p, q, r) : p, q, r \in \Lambda \text{ pairwise distinct}\}$ (the minimum exists since Λ is discrete).

Theorem 3. *For every lattice $\Lambda \subset \mathbb{R}^2$ of covolume 1, $P(\Lambda) \leq P_*$, with equality if and only if Λ is isometric to Λ_* .*

2. PROOF OF THEOREM 3

Lemma 4 (reduced basis). *After a rotation and a reflection, every unit-covolume $\Lambda \subset \mathbb{R}^2$ has a basis $u = (a, 0)$, $v = (p, a^{-1})$ with $a = \lambda_1(\Lambda)$, $0 \leq p \leq a/2$ and $|v| \geq a$. Moreover $a^4 \leq 4/3$.*

Proof. Lagrange–Gauss reduction; the bound is Hermite’s constant $\gamma_2 = 2/\sqrt{3}$. $\square$

Lemma 5 (upper bound). *With the notation of Lemma 4, $P(\Lambda) \leq G(a) := \min\{2a^3, \sqrt{a^2 + a^{-2}}\}$, and $\pi(0, u, v) \leq \sqrt{a^2 + a^{-2}}$ with equality iff $p = 0$.*

Proof. The collinear triple $(0, u, 2u)$ gives $\pi = a \cdot a \cdot 2a = 2a^3$. For the triple $(0, u, v)$, $\pi(0, u, v)^2 = a^2(p^2 + k)((a - p)^2 + k)$ with $k = a^{-2}$. Put $m = p(a - p) \in [0, a^2/4]$. Since $p^2 + (a - p)^2 = a^2 - 2m$,

$$(p^2 + k)((a - p)^2 + k) = m^2 + k(a^2 - 2m) + k^2 = (ka^2 + k^2) + m(m - 2k).$$

Because $a^4 \leq 4/3 < 8$ we have $m \leq a^2/4 < 2a^{-2} = 2k$, hence $m(m - 2k) \leq 0$ with equality iff $m = 0$, i.e. $p = 0$ (note $p \leq a/2 < a$). Therefore $\pi(0, u, v)^2 \leq a^2(ka^2 + k^2) = a^2 + a^{-2}$. $\square$

Lemma 6. *For $0 < a \leq (4/3)^{1/4}$ we have $G(a) \leq P_*$, with equality iff $a = a_*$.*

Proof. a_* is the positive root of $4a^8 = a^4 + 1$, equivalently $2a_*^3 = \sqrt{a_*^2 + a_*^{-2}}$. For $a \leq a_*$, $G(a) \leq 2a^3 \leq 2a_*^3$ with equality iff $a = a_*$. The function $g(a) = a^2 + a^{-2}$ is strictly decreasing on $(0, 1]$ and increasing on $[1, \infty)$. For $a_* < a \leq 1$, $G(a) \leq \sqrt{g(a)} < \sqrt{g(a_*)} = P_*$. For $1 \leq a \leq (4/3)^{1/4}$, $g(a) \leq g((4/3)^{1/4}) = \frac{2}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{7}{2\sqrt{3}}$, and $(\frac{7}{2\sqrt{3}})^2 = \frac{49}{12} < 16(\frac{1+\sqrt{17}}{8})^3 = P_*^4$ (numerically $4.083 < 4.202$), so again $G(a) < P_*$. $\square$

Lemma 7 (the extremal lattice). $P(\Lambda_*) = P_*$.

Proof. Write $u = (a_*, 0)$, $v = (0, a_*^{-1})$; note $a_* < 1$. The triples $(0, u, 2u)$ and $(0, u, v)$ give $\pi = P_*$, so $P(\Lambda_*) \leq P_*$. *Collinear triples.* Three distinct lattice points on a line are $q + jsw$, $q + ksw$, $q + lsw$ with w a unit vector, $s \geq \lambda_1 = a_*$ and integers $j < k < l$; then $\pi = s^3(k - j)(l - k)(l - j) \geq 2s^3 \geq 2a_*^3 = P_*$. *Non-collinear triples.* Let the side lengths be $d_1 \leq d_2 \leq d_3$ and γ the angle between the sides d_1, d_2 . The triangle has area $\geq 1/2$, so $d_1 d_2 \geq d_1 d_2 \sin \gamma = 2 \text{ area} \geq 1$ and $\pi \geq d_3$. We claim $d_3 \geq D := \sqrt{a_*^2 + a_*^{-2}} = P_*$. The three side vectors are nonzero, pairwise linearly independent vectors of Λ_* . A vector (ma_*, na_*^{-1}) has norm $< D$ only if $(m, n) \in \{(\pm 1, 0), (0, \pm 1)\}$: if $n \neq 0$ then $m^2 a_*^2 < a_*^2 + (1 - n^2) a_*^{-2} \leq a_*^2$ forces $m = 0$ and $n^2 < 1 + a_*^4 < 2$; if $n = 0$ then $m^2 < 1 + a_*^{-4} = 1 + y_\infty^2 < 4$. These vectors span only two directions, so at most two of the three pairwise independent side vectors can be shorter than D . Hence $d_3 \geq D$ and $\pi \geq P_*$. $\square$

Proof of Theorem 3. By Lemmas 4–6, $P(\Lambda) \leq G(a) \leq P_*$. If $P(\Lambda) = P_*$, then $G(a) = P_*$, so $a = a_*$; moreover $\pi(0, u, v) \geq P(\Lambda) = P_* = \sqrt{a_*^2 + a_*^{-2}}$, which by Lemma 5 forces $p = 0$, i.e. $\Lambda = \Lambda_*$. Conversely $P(\Lambda_*) = P_*$ by Lemma 7. $\square$

3. PROOF OF THEOREM 1

Let $\mathcal{L}_2$ be the space of unit-covolume lattices modulo isometries, with the topology induced by bases.

Lemma 8. (a) $T_\nu(\Lambda) \geq P(\Lambda)^{-\nu}$ for all Λ and $\nu > 4/3$. (b) P is continuous on $\mathcal{L}_2$ and $P(\Lambda) \leq 2\lambda_1(\Lambda)^3$. (c) T_ν is lower semicontinuous on $\mathcal{L}_2$. (d) For $\nu \geq \nu_0 > 4/3$: $T_\nu(\Lambda_*) \leq C P_*^{-\nu}$ with $C = T_{\nu_0}(\Lambda_*) P_*^{\nu_0}$.

Proof. (a) Translating a minimal triple so that one vertex is 0 produces a term $P(\Lambda)^{-\nu}$ of (1). (b) The bound is the triple $(0, u, 2u)$. Continuity: near a given lattice only triples of points in a fixed ball are relevant, and π is continuous in the basis. (c) T_ν is a sum of nonnegative continuous functions of the basis. (d) Every term satisfies $\pi \geq P_*$, so $(P_*/\pi)^\nu \leq (P_*/\pi)^{\nu_0}$. $\square$

Proof of Theorem 1. Existence. By Lemma 8(a),(b), $T_\nu(\Lambda) \geq (2\lambda_1^3)^{-\nu}$, so sublevel sets of T_ν lie in $\{\lambda_1 \geq \varepsilon\}$, which is compact by Mahler's criterion; with (c) a minimiser exists. *Convergence.* For a minimiser Λ_ν and $\nu \geq \nu_0$, Lemma 8 gives $P(\Lambda_\nu)^{-\nu} \leq T_\nu(\Lambda_\nu) \leq T_\nu(\Lambda_*) \leq C P_*^{-\nu}$, hence $P_* \geq P(\Lambda_\nu) \geq C^{-1/\nu} P_* \rightarrow P_*$. In particular $\lambda_1(\Lambda_\nu)$ is bounded below, so (Λ_ν) lies in a compact set. Any limit point Λ_∞ satisfies $P(\Lambda_\infty) = P_*$ by continuity, hence $\Lambda_\infty \cong \Lambda_*$ by Theorem 3. The same chain of inequalities gives $(\min T_\nu)^{1/\nu} \rightarrow 1/P_*$. $\square$

4. RELATION TO MODULAR GRAPH FUNCTIONS

For τ in the upper half plane the two-loop modular graph function is $C_{a,b,c}(\tau) = \sum_{p_1+p_2+p_3=0, p_i \neq 0} \prod_i \left(\frac{\text{Im } \tau}{\pi |p_i|^2}\right)^{c_i}$ $p_i \in \mathbb{Z} + \tau\mathbb{Z}$ [10]. With $\Lambda_\tau = (\mathbb{Z} + \tau\mathbb{Z})/\sqrt{\text{Im } \tau}$ (unit covolume) one has

$$C_{s,s,s}(\tau) = \pi^{-3s} T_{2s}(\Lambda_\tau),$$

so Theorem 1 is a statement about the location of the minimum of $C_{s,s,s}$ on the fundamental domain. At $s = 1$, Zagier's identity $C_{1,1,1} = E_3 + \zeta(3)$ and the minimality of the Epstein zeta function at ρ [1, 2, 3, 4] show that the minimum is at ρ (hexagonal). Theorem 1 shows that for large s it is near iy_∞ . We are not aware of earlier results on the minima of $C_{s,s,s}$ for $s > 1$; see the caveat in Section 8.

5. FINITE ν : THE MINIMUM OF $C_{s,s,s}$ FOR $s = 2, 3, 4$

Theorem 9 (computer-assisted). *For $s \in \{2, 3, 4\}$ the modular graph function $C_{s,s,s}$ attains its minimum over the fundamental domain only at $\tau = i$. Equivalently, for $\nu \in \{4, 6, 8\}$ the square lattice is the unique minimiser of T_ν among unit-covolume lattices in $\mathbb{R}^2$. Together with the case $s = 1$ (minimum at ρ , Section 4) and Theorem 1, the minimiser of $C_{s,s,s}$ is ρ for $s = 1$, i for $s = 2, 3, 4$, and tends to iy_∞ as $s \rightarrow \infty$.*

A universal majorant. Write $\tau = x + iy$, $\ell_v = |m + n\tau|^2/y$ for $v = (m, n)$, so that $T_\nu = \sum \prod_{\text{sides}} \ell^{-s}$. Around (x_0, y_0) put $A = a/y_0$, $B = b/y_0$. A short computation gives

$$\frac{\ell_v(x_0 + a, y_0 + b)}{\ell_v(x_0, y_0)} = 1 + \frac{(2\alpha^2 - 1)B + 2\alpha\beta A + \alpha^2(A^2 + B^2)}{1 + B},$$

with $\alpha^2 + \beta^2 = 1$ depending on v . Hence, coefficientwise, $\ell/\ell_0 - 1 \ll \hat{\varepsilon} := (A + B + A^2 + B^2)/(1 - B)$ and every triangle term t satisfies $t/t_0 \ll (1 - \hat{\varepsilon})^{-3s}$, uniformly in the lattice vectors. Consequently all Taylor coefficients of T_ν at τ_0 are bounded by $T_\nu(\tau_0)$ times the coefficients $\hat{G}_{ij}$ of $(1 - \hat{\varepsilon})^{-3s}$; this yields explicit remainder bounds and explicit bounds for the contribution of the terms omitted by a finite truncation.

Proof outline. (1) For $y > Y_0$ the collinear triples along the vector 1 give $T_\nu(\tau) \geq y^{3s} \sum_{j \neq k} (|j||k||j - k|)^{-2s} > T_\nu(i)$. (2) Local lemma: the symmetries $x \mapsto -x$ and $\tau \mapsto 1/\bar{\tau}$ fix i and force $c_{10} = c_{01} = c_{11} = c_{30} = c_{12} = 0$ in the Taylor expansion at i ; with rigorous enclosures of $c_{20}, c_{02}, c_{21}, c_{03}$ and the majorant remainder, $T(i + \delta) - T(i) > 0$ for $0 < \max(|A|, |B|) \leq r_{\text{loc}}$. (3) The rest of $[0, \frac{1}{2}] \times [0.8, Y_0]$ is covered by boxes on which the second-order Taylor polynomial at the centre (with rigorous coefficient enclosures) minus the majorant remainder exceeds an

upper bound for $T(i)$. Coefficients are computed from cube-truncated sums by direct (non-FFT) convolution with an explicit tail bound and a rounding margin.

Data of the three runs (scripts `theorem_mgf/prove_square_min.py`):

ν	$T_\nu(i)$ (enclosure)	$c_{20} \geq$	$c_{02} \geq$	r_{loc}	boxes
4	[7.5826697721, 7.5826700248]	0.43688	9.75784	0.00309	531
5	[4.8380268461, 4.8380268473]	2.58218	6.30394	0.00701	303
6	[3.2499743121, 3.2499743128]	4.27194	3.57092	0.00755	307
7	[2.2334461746, 2.2334461750]	5.34452	1.66169	0.00495	323
8	[1.5524012124, 1.5524012128]	5.83081	0.46080	0.00260	385

The runs for $\nu = 5, 7$ show that the square lattice is also the unique minimiser of T_5 and T_7 . The same approach failed for the hexagonal lattice at $\nu = 3$: close to $\nu = d$ the truncation tail decays only like R^{-4} , and at $R = 40$ the error exceeds the energy gap next to ρ . The small coefficient c_{02} at $\nu = 8$ reflects the nearby loss of stability of the square lattice at $\nu_2 \approx 8.606$. The status of the floating-point parts is the same as in Remark after Theorem 11: sums are sequential double-precision sums with explicit margins, not interval arithmetic.

6. NUMERICAL PHASE DIAGRAM FOR FINITE ν

The following are numerical observations, not theorems. T_ν was evaluated by cube-truncated direct summation with an FFT convolution and Richardson extrapolation in the cut-off (independent of [9]); agreement with the library of [9] is better than $2 \cdot 10^{-10}$ relative on 55 test cases, and the values at $\nu = 3$ reproduce those of [7] after conversion to unit covolume.

- $2 < \nu < \nu^*$: the hexagonal lattice is the global minimiser. At $\nu = 2$ this is exact by Section 4.
- $\nu^* = 3.9183649026 \dots$: first-order transition to the square lattice; both are local minima for $3.806 < \nu < 4.278$. *Certified*: $T_\nu(\mathbb{Z}^2) - T_\nu(\text{hex})$ changes sign in $(3.918364, 3.918366)$; this uses a direct (non-FFT) evaluation of the cube-truncated sum at $R = 100$, an explicit tail bound, and a rounding margin (script `certify_nustar.py`).
- $\nu_2 \approx 8.606$: the square lattice becomes unstable towards rectangular deformations; the optimal aspect ratio then grows continuously (1.056 at $\nu = 9$, 1.100 at 10, 1.141 at 12, 1.170 at 15, 1.194 at 20, 1.215 at 30), consistent with the limit $y_\infty = 1.2496$ of Theorem 1.

Conjecture 10. For $d = 2$ the minimiser of T_ν is hexagonal for $4/3 < \nu < \nu^*$, square for $\nu^* < \nu \leq \nu_2$ and rectangular with aspect ratio increasing from 1 to y_∞ for $\nu > \nu_2$.

7. THREE DIMENSIONS

For unit-covolume $\Lambda \subset \mathbb{R}^3$ the energy T_ν converges for $\nu > 2$, and the argument of Theorem 1 applies verbatim once the max-min problem is solved. For the body-centred cubic lattice $P(\text{BCC}) = 3/2$, attained by the isosceles triangles with two nearest-neighbour sides and one second-neighbour side, while $P(\text{FCC}) = P(\text{SC}) = \sqrt{2}$ (FCC: equilateral nearest-neighbour triangles; SC: right isosceles triangles).

Theorem 11 (computer-assisted). *For every lattice $\Lambda \subset \mathbb{R}^3$ of covolume 1, $P(\Lambda) \leq 3/2$, with equality if and only if Λ is isometric to the BCC lattice of covolume 1. Consequently, for $\nu > 2$ the energy T_ν has minimisers, and every family of minimisers converges to BCC as $\nu \rightarrow \infty$, with $(\min T_\nu)^{1/\nu} \rightarrow 2/3$.*

Scale-invariant formulation. Write $G = B^\top B$ for the Gram matrix of a basis B and $Q(G) = P(\Lambda)/\sqrt{\det G}$. By scaling we may assume $G_{11} = \lambda_1^2 = 1$ and use the parameters $q = (g_{22}, g_{33}, g_{12}, g_{13}, g_{23})$.

Lemma 12 (compact search region). *Every lattice has, up to isometry and scaling, a Gram matrix in the region $\mathcal{R}$ defined by $1 \leq g_{22} \leq g_{33}$, $0 \leq g_{12}, g_{13} \leq \frac{1}{2}$, $|g_{23}| \leq g_{22}/2$, $1 + g_{22} + 2(\varepsilon_1 \varepsilon_2 g_{12} + \varepsilon_1 g_{13} + \varepsilon_2 g_{23}) \geq 0$ for $\varepsilon_i = \pm 1$. On $\mathcal{R}$ one has $\det G \geq g_{22}g_{33}/4$, hence $Q(G) < 3/2$ whenever $g_{22}g_{33} > 64/9$.*

Proof. Take a Minkowski-reduced basis and flip the signs of b_2, b_3 to make $g_{12}, g_{13} \geq 0$; the listed inequalities are part of Minkowski's conditions. For the determinant: the triangle $(0, b_1, b_2)$ is acute, so the covering radius ρ of the plane lattice $L_2 = \mathbb{Z}b_1 + \mathbb{Z}b_2$ is its circumradius, $\rho^2 = g_{22}|b_2 - b_1|^2/(4\det_2)$ with $\det_2 = g_{22} - g_{12}^2 \geq \frac{3}{4}g_{22}$, whence $\rho^2 \leq (1 + g_{22})/3$. Minkowski reduction implies that the projection of b_3 to $\text{span}(b_1, b_2)$ lies in the Voronoi cell of L_2 , so the height h of b_3 above that plane satisfies $h^2 \geq g_{33} - \rho^2 \geq g_{33}/3$. Thus $\det G = \det_2 h^2 \geq g_{22}g_{33}/4$. Finally the collinear triple $(0, b_1, 2b_1)$ gives $Q \leq 2/\sqrt{\det G} \leq 4/\sqrt{g_{22}g_{33}}$. $\square$

In particular it suffices to consider the compact box $g_{22} \in [1, 8/3]$, $g_{33} \in [1, 64/9]$, $g_{12}, g_{13} \in [0, 1/2]$, $|g_{23}| \leq 4/3$. Within $\mathcal{R}$ BCC has exactly one representative, $q_0 = (1, 1, \frac{1}{3}, \frac{1}{3}, -\frac{1}{3})$ (checked by enumerating unimodular changes of basis).

Lemma 13 (local lemma, interval arithmetic). *Let $f_t(q) = \sqrt{\ell_m \ell_n \ell_{m-n}}/\det G$ for a triangle $t = (0, m, n)$, where $\ell_v = v^T G v$. At q_0 exactly 36 triangles (up to order) are active, $f_t(q_0) = 3/2$, and their gradients have 6 distinct values whose convex hull contains the origin at distance $c = 0.13887\dots$ from its boundary. On the box $q_0 + [-r, r]^5$ with $r = 0.007$, interval arithmetic gives $\|\nabla^2 f_t\|_F \leq M = 14.92$ for all active t . Hence $\min_t f_t(q_0 + \delta) \leq \frac{3}{2} - c|\delta| + \frac{M}{2}|\delta|^2 < \frac{3}{2}$ for $0 < |\delta| \leq \sqrt{5}r = 0.01565 < 2c/M = 0.01862$.*

Branch and bound. On a box B of parameters, each ℓ_v is linear in q and $\det G$ is a cubic polynomial; we bound $Q \leq \min_{t \in S} (\prod \max_B \ell)^{1/2} / (\min_B \det G)^{1/2}$ for a set S of candidate triangles (all $(0, m, n)$ with $m, n \in \{-1, 0, 1\}^3$, and the collinear $(0, m, 2m)$), with a relative safety margin 10^{-9} for rounding. Starting from the compact box above, boxes are discarded when they violate a region constraint, lie inside $q_0 + [-r, r]^5$, or have upper bound $< 3/2$, and are bisected otherwise. The run terminates after $9.93 \cdot 10^6$ boxes: *every* box is certified. Together with Lemmas 12 and 13 this gives $Q \leq 3/2$ on $\mathcal{R}$ with equality only at q_0 , i.e. Theorem 11. The convergence statement follows as in the proof of Theorem 1.

Remark 14 (status of the computer-assisted parts). The local lemma uses rigorous interval arithmetic (`mpmath.iv`). The branch and bound uses IEEE double precision with explicit safety margins (10^{-9} relative, 10^{-12} absolute on the determinant) that exceed the rounding errors of the few arithmetic operations per bound by several orders of magnitude; a fully interval-arithmetic re-run is straightforward and is planned. As sanity checks, the box bounds were compared with brute-force values of Q at 1200 random points (never violated), and the prover correctly fails when asked to certify $Q < 1.49$.

Finite ν in three dimensions (numerical). BCC is the best lattice found by a global search over all Bravais lattices for every tested $\nu \in [3.2, 12]$. FCC is a local minimum only for small ν : its curvature along the Bain path changes sign at $\nu_c = 3.7521 \pm 0.0001$ (spread over finite-difference steps and cut-offs). Among non-Bravais structures, hcp (with optimised c/a) lies slightly above FCC for all tested ν ; for $\nu \rightarrow \infty$ this is also clear from $P(\text{hcp}) = P(\text{FCC}) = \sqrt{2} < 3/2$. For mixed energies $E_\lambda = Z_\nu + \lambda T_\nu$ (pair Riesz energy $Z_\nu = \sum_{x \neq 0} |x|^{-\nu}$ plus three-body term, unit density) the optimum among FCC, BCC, hcp(c/a) and the Bain family jumps directly from FCC to BCC at a critical ratio $\lambda_c(\nu)$ (e.g. $\lambda_c \approx 0.05, 0.07, 0.11, 0.32$ for $\nu = 3.5, 4.5, 6, 9$); hcp is never optimal.

8. LIMITATIONS

(i) T_ν is a model energy; physically relevant three-body dispersion has an angular factor and is always accompanied by pair terms. (ii) The theorems concern Bravais lattices; hcp was only compared numerically. (iii) Apart from Theorems 1, 3, 11, 9 and the certified enclosure of ν^* , all statements are numerical; in particular the finite- ν phase diagrams are conjectural. (iv) The computer-assisted parts use double precision with explicit safety margins rather than interval arithmetic throughout. (v) Our literature search for prior results on minima of $C_{s,s,s}$ and on minimisers of T_ν was not exhaustive.

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